

RAHUL AGARWAL

AI & ML Engineer

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EDUCATION

Vellore Institute Of Technology, Bhopal

2023 – 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.84/10

TECHNICAL SKILLS

Languages: Python, Java

AI, ML & Computer Vision: Keras, YOLO, OpenCV, Albumentations, NumPy, Pandas

LLM & GenAI: Prompt Engineering, RAG, LLM APIs, Bedrock

Backend & Data Systems: FastAPI, SQLAlchemy, MySQL, Redis, Celery, Playwright, BeautifulSoup

Cloud & DevOps: AWS (EC2, S3, SageMaker), Docker

EXPERIENCE

• RS Mech Tech Pvt. Ltd.

Ahmedabad, India

Research Associate Intern

May 2026 – July 2026

- **AI & Computer Vision:** Built 4-stage ML workflows—data preprocessing, training, inference, and evaluation—using **Python, Keras, YOLO, OpenCV, and NumPy**.
- **Cloud Deployment:** Supported model storage and deployment using 3 AWS services—**S3, EC2, and SageMaker**—while documenting experiments and collaborating on solution improvements.

• Mantra Softech Pvt. Ltd.

Ahmedabad, India

Computer Vision Intern

May 2025 – June 2025

- **Vision Pipelines:** Developed computer vision pipelines for 2 core tasks—object detection and recognition—using **Python, YOLO, TensorFlow, and OpenCV**.
- **Model Optimization:** Optimized inference and deployment across 3 stages—preprocessing, detection, and post-processing—to improve real-world model performance.

PROJECTS

Lead Generation & AI Outreach Platform

06/2026

Python, FastAPI, SQLAlchemy, Playwright, BeautifulSoup, Celery, Redis, SMTP

Link: tinyurl.com/3ehev4mu

- Built an automated lead collection platform using Playwright and BeautifulSoup to scrape, clean, and structure **7 core business metrics** into a relational database via SQLAlchemy.
- Integrated Grok API for dynamic email personalization and configured asynchronous queue processing via Celery and Redis to handle resilient, high-volume batch delivery.
- Engineered an interactive Streamlit operational dashboard providing real-time telemetry, campaign analytics, delivery logs, and error diagnostic metrics for active outreach runs.

Real-Time Garbage Detection System

08/2025

Python, YOLOv11, OpenCV, Roboflow, Albumentations, MakeSenseAI

Link: tinyurl.com/59h5eyvs

- Annotated and augmented **1000+ waste images** with Albumentations and MakeSenseAI to train a custom YOLOv11 model for accurate multi-class classification.
- Developed a low-latency OpenCV video inference pipeline, reaching **86.4% mAP@50** and maintaining real-time processing speeds of **60 FPS** on the target system.
- Integrated the detection pipeline with a responsive web dashboard to stream live bounding-box telemetry and instant alert visual feedback.

CERTIFICATIONS

- [AWS Certified AI Practitioner](#) (AIF-C01) Amazon Web Services (AWS), 2026.
- [AWS Certified ML Engineer Associate](#) (MLA-C01) Amazon Web Services (AWS), 2026.

EXTRACURRICULAR

- Managed design, decor, and event logistics in Student Council, Meraki Club and Hindi Club.
- 2nd Runner Up in Badminton Tournament Hosted At Advitya Technocultural Fest.